

Lesson 6.12: Feed the Class, Menu Options and Constraints

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.3 Feed the Class project **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes **Room:** FACS lab with kitchen stations (no cooking today; a no-kitchen version is below)

Standards

- NYS module line: NYS FSP 5. Consumer Resources and Finance a) Demonstrate management of individual and family resources to select and purchase food
- NYS module line: NYS FSP 5. Consumer Resources and Finance b) Analyze decisions about providing safe and nutritious food for individuals and families
- NYS module line: NYS FSP 5. Consumer Resources and Finance c) Apply consumer skills to selecting and purchasing of food items
- NYS module line: NYS FSP 1. Basic Culinary Skills c) Demonstrate safety and sanitation procedures when handling food and tools (the constraint card; demonstrated on cook day in Lesson 6.16)
- CTE theme line: CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses
- CTE theme line: CTE PSI 2. Design Process (Proactive) a) c. Defining constraints that must be adhered to
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students are able to budget their time and money.
- CDOS Standard 3a Universal Foundation Skills: Managing Resources. Students understand the material, human, and financial resources needed to accomplish tasks and activities.
- National FCS 3.0: 2.1.3 Examine the decisions behind providing food that is safe and nourishing for individuals and families.
- National FCS 3.0: 8.4.2 Use menu-planning principles to build and adjust menus. (Confirmed in the 3.0 text in 02 Standards/STANDARDS SOURCES - web research.md, Area 8. It is not printed in the FSP module's own Standards Addressed block, so it is cited as a secondary competency.)

Enduring understanding and essential question

Big idea: A meal for a group is not a recipe. It is a budget, a set of limits, and a plan, and the limits are what make the plan hard. **Essential question:** Can we actually feed this whole class on what I can spend, in the time we have, with the rules this room has to follow?

Objectives

By the end of this lesson, students will be able to:

1. (Low) Given the constraint card, list the seven constraints on this project (no knives, no frying, pre-cooked meat only, the equipment available, the allergy list, 40 minutes, the station budget) and state what each one rules out.
2. (Mid) Given five costed menu option cards, compare two options on cost per person, difficulty, and equipment needed, and explain which one their station could actually finish inside a 40-minute period.
3. (High) Working as a station team, select one menu option, justify the choice against at least three constraints in writing, and assign all seven project roles.

Vocabulary

Tier 2 (general academic): constraint, criteria, justify, trade-off Tier 3 (FACS): food cost, per serving, station card, yield, allergen. See the unit vocabulary list for the Spanish column.

Materials and setup

- Handout, one per student plus one large copy per station: Handout 06.12 - Menu Option Cards and Constraint Card.md (the constraint card, five costed menu option cards, the station choice sheet, the role sheet)
- The class allergy list, printed and posted at the front and at every station.
- Four large sheets of paper or the whiteboard divided into four station areas
- A calculator per pair
- One real supermarket circular per station, this week's, for the do now only (they get used for real in Lesson 6.13).
- Slides: Slides 06.12 - Feed the Class Menu Options and Constraints (outline in 05 Materials; the site renders it as a deck)
- Setup notes: post the allergy list before the bell. Set the four station areas so teams sit where they will cook. No food is handled today, but the allergy list goes up today because every decision made in this period depends on it.

Pre-assessment

The Food Systems Quiz from Lesson 6.5 and the three labs in Topic 6.2 already told you who can read a recipe, who can hold a role, and which station needed the most reminders. Use that to set the four station teams before this period, not during it. The do now (guess the total) is a second read on whether students have any idea what food costs for a group.

Do now and hook (Time: 5 min)

On the board: "Look around this room. Write down what you think it would cost to feed everybody in here one real meal. One number. No talking."

Debrief: collect numbers from six students, low to high, and write them on the board. The spread is usually huge: \$20 to \$300. Then the hook: "Here is the real number. I can spend about \$20 per station, so about \$80 for this class. That is under three dollars a person. In two weeks you are going to plan it, price it, cook it, and serve it, and I am not going to do it for you. You are the kitchen."

Say the one thing that makes this land: “I ran this at my last school. Four kitchens in one period, every dish priced from the store circular, 112 students across four periods on \$150. It worked because the plan was on paper before anybody turned on a burner.”

Procedure

Step	Teacher will	Students will	Time
1. Direct instruction: the brief	Read the brief off the slide in four sentences: your station plans one dish, scales it for the class, prices it from a real circular, writes its own flowchart and station card, cooks it, and serves it. Name the five things that get turned in and say the rubric is in their hands today, not at the end. Hand out Handout 06.12 and Project 06 with Rubric 06.	Read the brief. Write their station number and their name on the handout.	5 min
2. Direct instruction: the constraint card	Walk the seven constraints one at a time. For each, ask the class what it rules out before you say it: no knives rules out chopping an onion; no frying rules out chicken cutlets and anything in a pan of oil; pre-cooked meat only rules out raw chicken and raw ground beef; the equipment list rules out anything needing a blender or a grill; the allergy list rules out named items for this class; 40	Fill the “what this rules out” column on the constraint card as each one is named. Read the posted allergy list and copy the allergens onto the card.	8 min

	minutes rules out anything that bakes for 45; the station budget rules out steak. Post the card at each station.		
3. Learning activity: the menu option walk	Put the five option cards on the five tables (or the five wall spots). Give teams 90 seconds per card. At each card they write three things on the choice sheet: the total cost for 28 portions, the cost per person, and one reason this dish could go wrong in 40 minutes. Ring the reset cue to rotate.	Rotate all five cards in teams. Fill the comparison table: cost, cost per person, difficulty mark, equipment, one risk.	9 min
4. Practice: the station choice and the justification	Give the rule: the class menu needs at least one main, at least one side or vegetable, and one dessert, so no two stations take the same option. Teams rank their top two on a slip; you resolve conflicts by the ranking and by which team can handle the equipment. Then teams write the four-sentence justification: one sentence on why they picked the dish, then one sentence per constraint, naming the constraint and how their dish clears it.	Rank the top two. Write the justification: "We picked . <i>It clears the no-knife rule because</i> . It fits 40 minutes because . <i>It comes in at</i> per person, which is under the station budget because ____."	8 min

	Note that Option B comes in \$1.32 over the station budget as written, and any team that picks it owes you a fix in Lesson 6.13.		
5. Roles and closure setup	Read the seven project roles off the slide and say what each one owns and which lesson it owns it in. Every role does something on every day; nobody holds a clipboard and watches. Teams fill the role sheet and hand it in.	Assign all seven roles by agreement, or by the team's own rule if they cannot agree. Write the roles on the sheet with names.	3 min

Do now 5 + steps 33 = 38, plus the 2-minute closure. If Step 4 runs long, the role sheet finishes as the do now of Lesson 6.13.

Questions to ask

Monitor understanding:

- Which constraint rules out a stir fry? Which one rules out a chicken cutlet?
- Option A costs \$18.31 for 28 portions. What is that per person? Show me the division.
- Your dish needs the oven. So does Station 4's. What is the problem, and who has to solve it?
- Where on your card does it say how many portions it makes? If it does not say, what do you do?

Deepen learning:

- I gave you the constraints. In a real restaurant, who sets them, and what happens if the kitchen ignores them?
- Two of these options are cheaper per person than the school lunch. Why do you think that is, and what is the school paying for that we are not?
- Your dish is the cheapest option on the table. Is cheapest the same as best here? What else matters to the people eating it?
- One of these five dishes has no heat at all. Is that a weaker choice or a smarter one? Say why with a reason a chef would give.

Check for understanding

Clipboard walk during Step 3 and Step 4, a yes or no per station on three things: the cost per person is computed correctly on the choice sheet, the risk column names a real risk and not “we might mess up,” and the justification names an actual constraint by its words. A station with three yeses gets its option confirmed at the end of the period. A station with a no on the justification gets one question from you at Step 5 and rewrites that sentence before the bell.

Closure (Time: 2 min)

Each station says one sentence out loud, in this form: “We are Station , ***we are making*** , and the constraint that will be hardest for us is ____.” Write the four hardest constraints on the board and leave them up; the flowchart in Lesson 6.14 has to solve them. Exit ticket is the role sheet handed in at the door.

Differentiation and supports

- ELL: every menu option card has a picture of the finished dish and a picture of each ingredient in its store package; the constraint card uses a symbol for each constraint (a crossed-out knife, a crossed-out pan of oil, a clock, a dollar sign) so the card can be read without the sentences; the words constraint, budget, portion, allergen, and per person are on the unit vocabulary card in Spanish; the justification has a sentence frame in English and Spanish on the back of the choice sheet.
- IEP and 504 (general): the comparison table can be filled with checkmarks and numbers only; a calculator for everyone, not just those who ask; the card walk can be done seated with the cards brought to the table for any student who cannot rotate; the role sheet lets a student name one role they want before the team negotiates, so a student who needs a specific job is not talked out of it.
- Grade 6 support: three option cards instead of five (Option B pizza tortillas, Option D garlic bread and vegetable tray, Option E dessert cups, which are the three easiest), and the justification is two sentences instead of three.
- Grade 8 stretch: the team also computes what the dish would cost per person if the class doubled to 56, and says which line on the list stops being a good deal at that size (the answer is usually the aerosol whipped topping and the small cheese bags: more small packages, no savings). A grade 8 team may also propose a sixth option of its own, priced from the circular, and pitch it to you for approval.
- UDL checkpoint used: multiple means of representation (each option is a picture, a price table, and a difficulty mark, so the same decision can be made three ways) and multiple means of engagement (the team picks its own dish and its own roles, and the whole class meal depends on the choice).
- No-kitchen alternative: the whole lesson is paper already. For a room with no kitchen, swap Option A (spaghetti, needs a stove) and Option B (pizza tortillas, needs an oven) for the two no-cook options on page 6 of the handout: a build-your-own sandwich or wrap bar, and a snack board with pre-cut fruit, crackers, cheese sticks, and hummus. The math, the flowchart, the shopping list, and the service are identical. Project 06 has the full no-kitchen version.

Allergy line

No food is handled today. The class allergy list goes up today anyway, because every dish chosen in this period has to clear it. A station whose chosen option contains an allergen on the list either swaps the item (dairy-free cheese, a nut-free spread) or swaps the option, and the swap is written on the choice sheet before the period ends.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. List the seven constraints and what each rules out	FSP 1. c); PSI 2. a) c.; CDOS 3a Managing Resources	Constraint card, Step 2	Formative; card checked for seven rows filled and a real exclusion in each
2. Compare two options on cost per person, difficulty, and equipment	FSP 5. a), 5. c); FCL 4. c); Std 3 PI C; National FCS 8.4.2	Option walk comparison table, Step 3	Formative; clipboard yes or no on the cost per person math and the risk column
3. Select an option, justify against three constraints, assign roles	FSP 5. a), 5. b), 5. c); PSI 2. a) c.; National FCS 2.1.3	Choice sheet and role sheet, Steps 4 and 5	Formative for daily work; the choice sheet is page 1 of the project folder and feeds Rubric 06 criterion 2

The whole project is scored summatively on Rubric 06 - Feed the Class.md at the end of Lesson 6.17. Topic 6.3's formal check is that rubric; there is no separate quiz for this topic.

Homework

None required. Optional and useful: ask at home what a dinner for your family costs and whether anybody knows the price of one thing in it. Bring the number, not the story.

Connections

Math: the cost per person division and the scale factor in Lesson 6.13 are ratio and unit rate work. Show the math teacher the option cards; if the grade 7 math class is on unit rate that month, the same five cards are a ready problem set. Community: the manager of the supermarket the circular comes from. FACS gives the store a class of customers who can read a unit price and a thank-you note; the store gives FACS this week's circular in a stack, a five-minute answer on why a store brand costs less, and sometimes a donation toward the \$80.

Sources

- NYS Middle Level CTE FACS Content Module 06, Food Systems and Production (nyc-tecenter.org, June 2018), lines 1. c), 5. a), 5. b), and 5. c). Copy in 02 Standards/NYS Module PDFs/.

- NYS Middle Level CTE Theme Module 03, Financial and Consumer Literacy, line 4. c); Theme Module 05, Problem Solving and Innovation, line 2. a) c. Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6.
- CDOS Intermediate Standard 3a, Managing Resources, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 2.1.3 and 8.4.2, per 02 Standards/STANDARDS SOURCES - web research.md.
- My own materials, from two runs. The run in the hook: 112 students across four periods and four kitchens on \$150. The second example, a different run: six costed menus for 165 students across six periods and three kitchens on a \$200 budget, with the standing constraints “No knives, No frying, Pre-cooked meat only, Stovetop plus oven”. The five option cards in Handout 06.12 are rescaled from the six menus of that second run for one class of 28 at four stations. Prices are my verified supermarket prices from the week of March 20 to 26, 2026, to be refreshed from the current circular before teaching.

Teacher notes

- If the period runs short, cut Step 3 to three cards (the two stations that will not use them see them in the handout) and never cut Step 2. A team that does not know the constraints writes a plan you have to throw away.
- Set the four station teams before the period. Mixing one strong reader, one strong measurer, and one student who keeps a group calm into each team is worth more than letting friends pick.
- Common mistake: a team computes cost per person by dividing by 7 (their station) instead of 28 (the class). The option card says “28 portions” in bold for that reason. Catch it at Step 3, not at Step 4.
- The over-budget option is on the table on purpose. A team that notices Option B is \$1.32 over before you say it is a team that will find the store-brand swap in Lesson 6.15 without help.
- Two stations will both want the oven. Let them discover it here. The fix is the flowchart in Lesson 6.14: one bakes in minutes 10 to 20, the other in minutes 20 to 30, or one moves to the toaster oven.
- Keep the four “hardest constraint” sentences on the board through Lesson 6.16. On cook day, point at them.